Vedant Desai

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EDUCATION

University of Michigan — Expected May 2028
B.S. in Computer Science and Economics — Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

CaseStudyPrep.AI — Dec 2025 – May 2026
Software Engineer Co-op (Voice AI) — Remote

- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.
- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.

Vylet — May 2026 – Present | vyletdata.com
Founder — Automated lead-sourcing platform for PE/search-fund; live product generating $1,500 MRR across three clients

- Launched Vylet, automating a 30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Diagnosed a name-collision defect in ownership-verification logic that was incorrectly rejecting valid acquisition targets sharing a name with an unrelated business elsewhere; the fix lifted the pipeline's lead-qualification rate from 79% to 89% with zero change to sourcing volume.
- Grew Vylet from 1 to 3 paying subscription clients within six weeks of launch — spanning workforce-software, landscaping, and geography-first sourcing engagements — generating $1,500 in monthly recurring revenue.

Michigan Data Consulting (MDC) — Jan 2026 – May 2026
Data Engineer — Michigan Campaign Finance Network — Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at 2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating 800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.
- Scoped technical delivery directly with MCFN stakeholders as the only engineer on contract — from ingestion pipelines through REST endpoints on EC2 — with no backend team to share infrastructure, API design, or deployment ownership.

PROJECTS

Granular Synthesizer Plugin | *C++, JUCE, VST3, AU* — github.com/Verdent06/granular-synth

- Enforced the audio thread's zero-allocation constraint by pre-allocating a MemoryPool<Grain, 64> slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so processBlock() never calls new or delete after prepareToPlay() completes.
- Wired UI-to-audio delivery through a hand-rolled SPSC FIFO (64-slot fixed array, atomic head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic shared_ptr swap so the UI thread never blocks processBlock().
- Implemented 16-voice polyphony with per-voice ADSR envelopes, pre-computing attack/decay/release rates as samples-per-step in prepareToPlay() to eliminate division inside the audio hot loop — and a voice-stealing algorithm that evicts the oldest active note by age counter when all 16 slots are busy, with each voice falling back to a sine oscillator until a WAV sample is loaded.

TECHNICAL SKILLS

Languages .... Python, TypeScript, C++, SQL, HTML/CSS
Frameworks .... Angular, Flask, RxJS, JUCE
Cloud & Infrastructure .... AWS (EC2, S3), Docker, Celery, Git